AP PRECALCULUS · TOPIC 1.9

Rational Functions and Vertical Asymptotes

Section 1 — Multiple Choice
Circle the letter of the correct answer

Name ..... Class _____ Date _____

No credit for unsupported answers in Section 2. Questions 1–8: 1 point each (8 points).

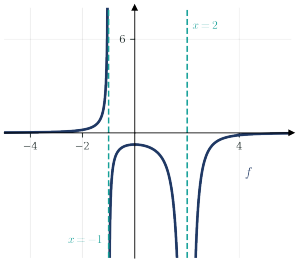
1 Which of the following gives all vertical asymptotes of $r(x) = \dfrac{x+4}{x^2-x-12}$?

A. $x = -3$ and $x = 4$
B. $x = 3$ and $x = -4$
C. $x = -4$
D. $x = -4$, $x = -3$ and $x = 4$

2 Which of the following gives all vertical asymptotes of $r(x) = \dfrac{x^2-9}{x^2-x-6}$?

A. $x = -2$ and $x = 3$
B. $x = -2$ only
C. $x = 3$ only
D. $x = -3$ and $x = 2$

3 The graph of the rational function f is shown, with vertical asymptotes at $x = -1$ and $x = 2$. Which statement is true?



A. $\lim\limits_{x\to -1^+} f(x) = \infty$ and $\lim\limits_{x\to 2^+} f(x) = -\infty$
B. $\lim\limits_{x\to -1^+} f(x) = -\infty$ and $\lim\limits_{x\to 2^+} f(x) = \infty$
C. $\lim\limits_{x\to -1^+} f(x) = -\infty$ and $\lim\limits_{x\to 2^+} f(x) = -\infty$
D. $\lim\limits_{x\to -1^+} f(x) = 0$ and $\lim\limits_{x\to 2^+} f(x) = 0$

4 What is the value of $\lim\limits_{x\to 3^-} \dfrac{x+1}{x-3}$?

A. ∞
B. 0
C. 4
D. $-\infty$

5 Let $g(x) = \dfrac{5}{(x+2)^2(x-1)}$. Which statement is true?

A. $\lim\limits_{x\to -2^-} g(x) = -\infty$ and $\lim\limits_{x\to -2^+} g(x) = \infty$
B. $\lim\limits_{x\to -2^-} g(x) = -\infty$ and $\lim\limits_{x\to -2^+} g(x) = -\infty$
C. $\lim\limits_{x\to -2^-} g(x) = \infty$ and $\lim\limits_{x\to -2^+} g(x) = \infty$
D. $\lim\limits_{x\to -2^-} g(x) = \infty$ and $\lim\limits_{x\to -2^+} g(x) = -\infty$

6 A rational function has vertical asymptotes at $x = -2$ and $x = 5$, a zero at $x = 1$, and no other zeros. Which could be its equation?

A. $r(x) = \dfrac{x+1}{(x-2)(x+5)}$
B. $r(x) = \dfrac{(x+2)(x-5)}{x-1}$
C. $r(x) = \dfrac{x-1}{(x+2)(x-5)}$
D. $r(x) = \dfrac{x-1}{(x-2)(x+5)}$

7 The cost, in thousands of dollars, of removing p percent of a pollutant is $C(p) = \dfrac{50p}{100-p}$ for $0 \le p < 100$. Which is the best interpretation of $\lim\limits_{p\to 100^-} C(p) = \infty$?

A. Removing 100% of the pollutant costs exactly 100 thousand dollars
B. The cost never exceeds 50 thousand dollars
C. The cost decreases toward 0 as the percentage removed approaches 100
D. As the percentage removed gets close to 100, the cost increases without bound

8 Which gives all vertical asymptotes of $r(x) = \dfrac{(x-2)^2(x+1)}{(x-2)^3(x+4)}$?

A. $x = -4$ and $x = 2$
B. $x = -4$ only
C. $x = -4$, $x = -1$ and $x = 2$
D. $x = 4$ and $x = -2$

Rational Functions and Vertical Asymptotes

Section 2 — Free Response
Show all working; answers need units and limit notation where relevant

Question 9 Calculator allowed · 6 points

For a thin lens with focal length 10 cm, an object placed u cm from the lens ($u > 10$) forms an image v cm from the lens, where $v(u) = \dfrac{10u}{u - 10}$.

(a) Find the equation of the vertical asymptote of the graph of v and justify your answer. DOK3 2 pt

(b) Write a limit statement describing the behavior of $v(u)$ as u approaches 10 from the right, and interpret it in context. DOK3 2 pt

(c) Find the object distance u for which the image distance is 25 cm. Give your answer to three decimal places. DOK3 2 pt

Question 10 No calculator · 6 points

Let $r(x) = \dfrac{x^2 - 2x - 3}{x^2 - 2x}$.

(a) Factor the numerator and the denominator of r. Give the zeros and the equations of all vertical asymptotes. DOK3 2 pt

(b) Find $\lim\limits_{x \to 2^-} r(x)$ and $\lim\limits_{x \to 2^+} r(x)$. Show the sign of each factor. DOK3 2 pt

(c) Write an equation for a rational function q with vertical asymptotes at $x = 0$ and $x = 2$ whose graph goes to the same infinity on both sides of $x = 0$. Justify your answer. DOK3 2 pt